

# **AI-Powered Nutrition Agent**

## **Problem Statement-:**

Many individuals struggle to maintain a healthy diet due to a lack of personalized nutritional guidance, busy lifestyles, and limited access to reliable dietary information. Generic diet plans often fail to consider a person's age, body measurements, health conditions, and fitness goals, resulting in ineffective nutrition management.

The objective of this project is to develop an AI-Powered Nutrition Agent that provides personalized diet recommendations, BMI analysis, nutrition information, and health advice based on user inputs. The system helps users make informed dietary choices, track their nutritional intake, and adopt healthier eating habits through an interactive and user-friendly platform.

By leveraging AI-driven recommendations and nutrition data analysis, the proposed solution aims to improve health awareness, support fitness goals, and promote a balanced lifestyle.

The solution also enables users to search for nutritional information about various foods, monitor their daily calorie and nutrient intake, and receive health advice relevant to their medical conditions. By integrating AI-driven recommendations and nutrition data analysis, the system assists users in achieving their health and fitness objectives while promoting healthier lifestyle habits.

## **AI Nutrition – Solution Overview**

AI Nutrition is an intelligent system designed to help users maintain a healthy diet by analyzing personal data and recommending customized meal plans.

### **Proposed Solution (AI-Based)**

The system solves this using AI by:

#### **1. User Input Collection**

- Age, weight, height, gender
- Activity level
- Health goal (weight loss / gain / maintenance)

#### **2. Data Processing**

- Calculates BMI (Body Mass Index)
- Determines daily calorie requirement (TDEE)

#### **3. AI Recommendation Engine**

- Suggests food items based on:
  - Calories
  - Protein, carbs, fats
  - Nutrition balance

#### **4. Diet Plan Generation**

- Breakfast, lunch, dinner plan
- Indian food options (customizable)

#### **5. Optional Features**

- Barcode/food scanning
- Chatbot diet assistant
- Water intake tracker

### **Outcome**

- Personalized diet plans
- Easy calorie tracking
- Better health management using AI